

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2023-2024
FULL MARKS: 75

CSE 4835: Pattern Recognition

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Justify your agreement/disagreement with the following statements: 3 × 4
(CO2)
(PO2)
 - i. Training on higher-level image features (e.g., color histograms or HOG features) instead of raw pixels improves image classification accuracy across all types of models.
 - ii. If the datapoints are linearly separable, then training a linear classifier with a Softmax loss and no regularization will achieve zero training loss at convergence.
 - iii. Given a constant gradient at each time step, the update to the weights will decrease over time when using the AdaGrad optimizer.
 - iv. Consider the Loss function given in Equation 1, where $R(w)$ is a L2 regularizer and L represents standard training loss. If we set $\lambda = -1$, then for every w , there exists another set of weights w' with lower loss, but an accuracy less than or equal to that of w .
$$L(w) = \frac{1}{N} \sum_{i=1}^N L_i(f(x_i, w), y_i) + \lambda R(w) \quad (1)$$
- b) Draw a computational graph for the function given in Equation 2. Calculate the values of each $\frac{df}{dw_1}$ and $\frac{df}{dw_2}$ considering $w_0 = 1, w_1 = 3, w_2 = 4, x_0 = 3, x_1 = 2, b_0 = 8, b_1 = 1$. 8
(CO1)
(PO1)

$$f(x_0, x_1, w_0, w_1, w_2, b_0, b_1) = \max(w_2 \times (w_0 \times x_0 + w_1 \times x_1 + b_0) + b_1, 0) \quad (2)$$
2. a) Explain the significance of using an activation function using the concept of space warping. 7
(CO1)
(PO1)
- b) 'A Neural Network with one hidden layer can approximate any function $f: \mathbb{R}^N \rightarrow \mathbb{R}^M$ with arbitrary precision' — Justify the statement. 10
(CO2)
(PO2)
- c) Discuss the properties of Adam optimizer that contribute to its widespread adoption as a default optimization algorithm across diverse downstream tasks. 8
(CO1)
(PO1)
3. a) i. ResNet-50 represents an enhanced variant of ResNet-34, achieving a 1.45% reduction in the Top-5 error rate on the ImageNet dataset, while incurring only an additional 0.2 GFLOPs of computational overhead. Discuss the architectural design choices that contributed to this performance improvement. 5
(CO2)
(PO2)
- ii. The VGG16 and VGG19 both have five convolutional stages. The VGG19 architecture has additional convolution layers in the later stages. Assume these additional layers are introduced in the earlier stages instead of later. Criticize whether this design choice will improve computational efficiency and/or classification performance. 5
(CO2)
(PO2)

- b) Consider a 3D convolutional network with its layers specified in Table 1. For each layer, fill the blank fields in the table with the size of the activation volumes, and the number of weight and bias parameters. The INPUT layer is provided; this network takes in 8 frames of 64-by-64 pixel RGB images. Write your answer as a multiplication (e.g. $128 \times 128 \times 128$). The layer descriptions are as follows:

- CONV3-N-p1-s2 is a convolutional layer with $N \times 3 \times 3$ kernels, padding 1 and stride 2 (in height, width, and temporal dimension) with ReLU activation.
- POOL2 denotes a $2 \times 2 \times 2$ max-pooling layer, with stride 2 and no padding.
- FC-N denotes a fully-connected layer with N neurons, with ReLU activation.

Table 1: Layer description for Question 3.b

Layer	Activation Volume	Learnable Weight Parameters	Biases
INPUT	$3 \times 8 \times 64 \times 64$	N/A	N/A
CONV5-16-p2-s1			
POOL2			
CONV3-16-p1-s1			
POOL2			
CONV3-32-p1-s2			
FC-256			
FC-10			

- c) In a 3D convolutional network, we can define spatiotemporal batch normalization similarly to operate on a batch of 4D feature maps of shape $N \times C \times T \times H \times W$, normalizing over the N, T, H , and W dimensions. If we now add one spatiotemporal Batch Normalization (BN) after each convolution layer, how many additional parameters will be introduced in Table 2?

Table 2: Number of BN parameters for Question 3.c

Layer	BN Parameters
CONV5-16-p2-s1	
CONV3-16-p1-s1	
CONV3-32-p1-s2	

- d) Figure 1 shows an image of size 16×16 (partially shown). Design a 5×5 convolutional filter that is maximally activated when sliding over the '3' in Figure 1. Consider the black pixels as 1s, and the white pixels are -1. For simplicity, use only -1s and 1s in your designed filter.

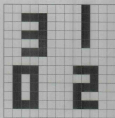


Figure 1: Sample image for Question 3.d